Worksheet 24

December 2, 2026 Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

1. Review: half-angle signs and values

Suppose $\cos\theta = -\frac{5}{13}$ and $\pi < \theta < \frac{3\pi}{2}$.

(a) Identify the quadrant of $\frac{\theta}{2}$.

(b) Find $\sin\frac{\theta}{2}$ and $\cos\frac{\theta}{2}$ exactly. Simplify the radicals.

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December 2, 2026 Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

2. Review: inverse values and equations

Give exact values and show your work.

(a) Evaluate $\arccos\left(\cos\frac{5\pi}{4}\right)$ using the principal range.

(b) Solve $2\sin^2\theta + \sin\theta = 0$ on $0 \le \theta < 2\pi$.